

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

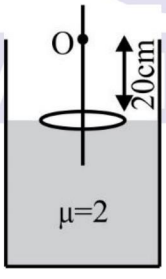
Sec: SR STAR - A & APEX, ALL IITS **DT: 12/01/2026**
Time: 3 Hrs **GTM - 16** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

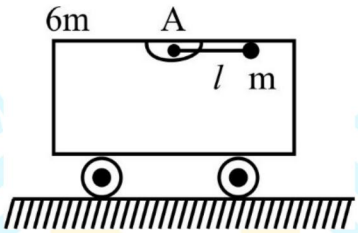
1. A point object O is placed at a distance of $20\ \text{cm}$ in front of an equi-convex lens $(\mu = 1.5)$ of focal length $10\ \text{cm}$. The lens is placed on a liquid of refractive index 2, as shown. Image will be formed at a distance h from lens. The value of h is



A) 5 cm B) 10 cm C) 20 cm D) 40 cm

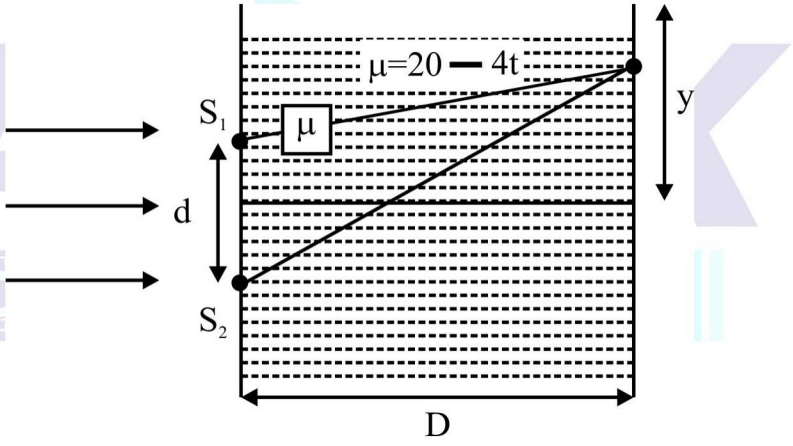
For More Join: @IITJEE_Advanc

2. In the figure shown the cart of mass $6m$ is initially at rest. A particle of mass m is attached to the end of the light rod which can rotate freely about A. If the rod is released from rest in a horizontal position shown, determine the velocity v_{rel} of the particle with respect to the cart when the rod is vertical



A) $\sqrt{\frac{7}{3}gl}$ B) $\sqrt{\frac{7}{6}gl}$ C) $\sqrt{\frac{14}{3}gl}$ D) $\sqrt{\frac{8}{3}gl}$

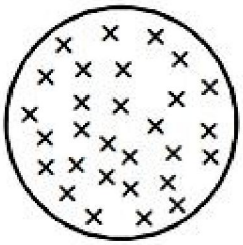
3. A parallel beam of monochromatic light of wavelength $\lambda = 100(A^\circ)$ is incident on the slits separated by distance $d = 2\ mm$. There is a screen at a distance $D = 1m$ from slit. R.I of the medium between slits and screen is varying with time as $\mu = 20 - 4t$ until it becomes 1. A glass slab of R.I. $\mu = 5$ and thickness $0.2\ mm$ is placed infront of one of the slit S_1 as shown in figure. In figure, y represent position of central maxima on the screen from its geometrical centre. Then calculate value of y (in cm) at $t = 5\sec$.



A) 40 B) 7.5 C) 20 D) 15

For More Join: @IITJEE_Advanc

4. A uniform magnetic field $B = kt$ is passing perpendicular to plane of a ring of resistance R. k is a constant and t is time. If resistance of the ring is constant, the current flowing through the ring is



A) Clockwise and constant

B) Anticlockwise and constant

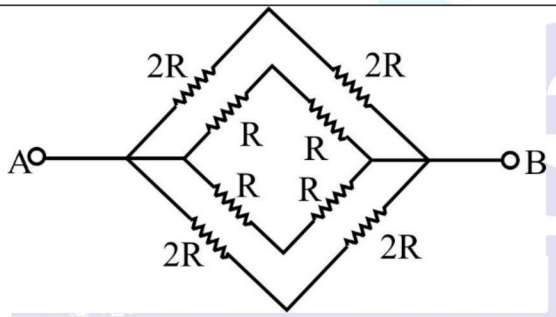
C) Zero

D) Anticlockwise and increasing with time

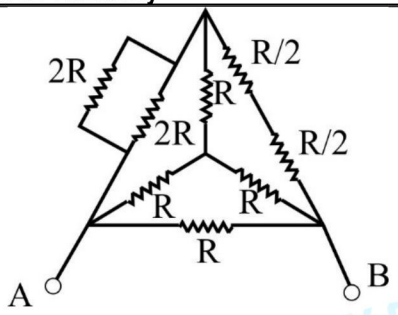
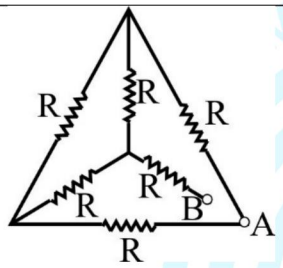
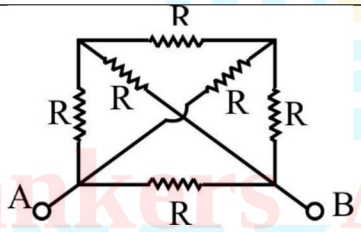
5. A prism having refractive index $\sqrt{2}$ and refracting angle $30°$ placed in air, has one of the refracting surfaces polished. A beam of light incident on the other refracting surface will retrace its path if the angle of incidence is

A) $0°$ B) $30°$ C) $45°$ D) $60°$

6. List – I shows electric circuits and List – II shows equivalent resistance between A and B. Match List – I with List – II.

	List – I		List – II
P)		1.	$\frac{R}{2}$

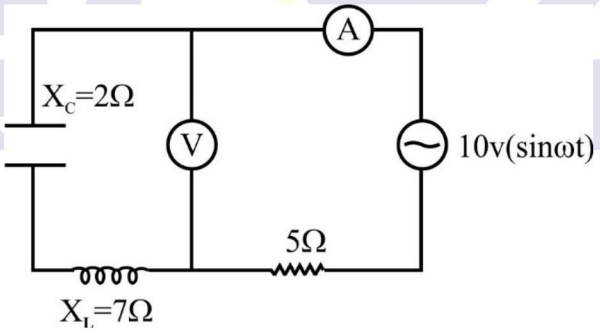
For More Join: @IITJEE_Advanc

Q)		2.	$\frac{R}{3}$
R)		3.	$\frac{2R}{3}$
S)		4.	$2R$

A) $P \to 3; Q \to 1; R \to 4; S \to 2$ B) $P \to 2; Q \to 1; R \to 4; S \to 2$

C) $P \to 3; Q \to 1; R \to 4; S \to 1$ D) $P \to 2; Q \to 1; R \to 4; S \to 3$

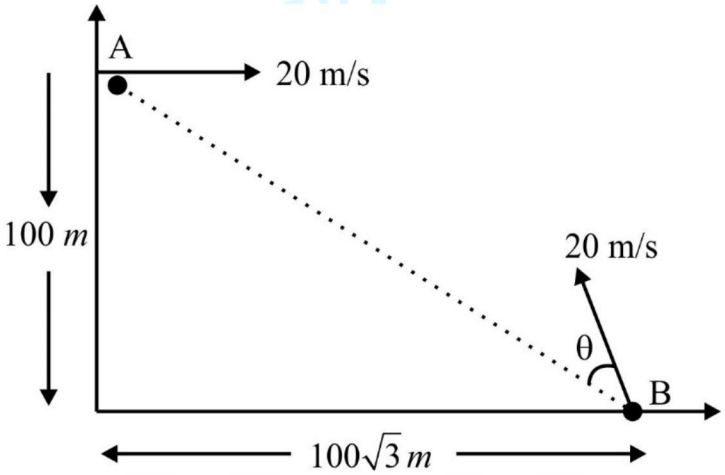
7. In the figure shown hot wire voltmeter and hot wire ammeter are ideal. The reading of voltmeter is



A) $5\sqrt{2}V$ B) $5V$ C) $10V$ D) None of these

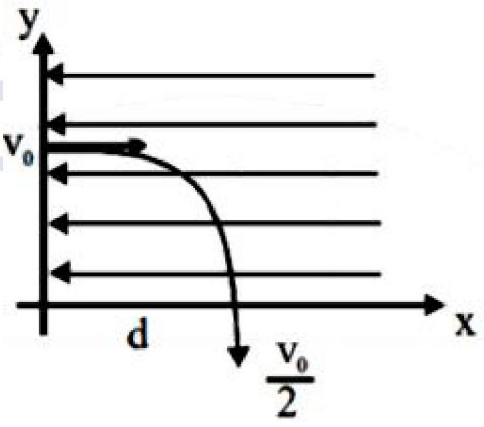
For More Join: @IITJEE_Advanc

8. Two small balls A and B are launched in the same vertical plane simultaneously, with same speed of $20\ \text{m/s}$ at $t=0$. Ball A has a horizontal initial velocity and ball B has initial velocity at an angle θ above the line joining A and B as shown. If the projectiles collide in mid-air at some time, then



A) $\theta=45°$ B) $\theta=60°$ C) $\theta=30°$ D) $\theta=37°$

9. In a certain region, free from gravity, electric field is along negative x-direction and it is constant. A particle having mass 'm' and charge q is projected along x-direction with speed v_0. An additional force $\vec{F}=\vec{C}\times\vec{v}$ is acting on the charge where $\vec{v}$ is velocity vector and $\vec{C}$ is a constant vector. The charge comes out of region with speed $\frac{v_0}{2}$. Then the magnitude of electric field will be

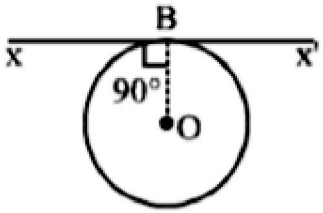


For More Join: @IITJEE_Advanc

A) $\frac{3}{4}\frac{mv_0^2}{qd}$ B) $\frac{4}{3}\frac{mv_0^2}{qd}$

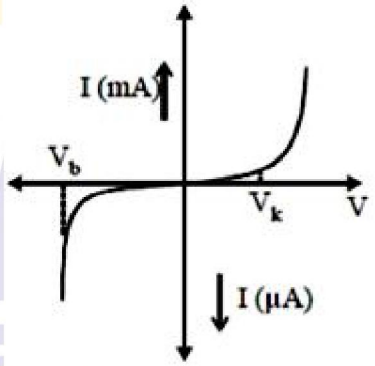
C) $\frac{3}{8}\frac{mv_0^2}{qd}$ D) cannot be determined

10. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is:



A) $\frac{\rho L^3}{8\pi^2}$ B) $\frac{\rho L^3}{16\pi^2}$ C) $\frac{5\rho L^3}{16\pi^2}$ D) $\frac{3\rho L^3}{8\pi^2}$

11. The $V-I$ characteristic for a $p-n$ junction diode is plotted as shown in the figure. From the plot we can conclude that [$V_b \rightarrow$ breakdown voltage, $V_k \rightarrow$ knee voltage]



A) the forward bias resistance of diode is very high; almost infinity for small values of V and after a certain value it becomes very low.

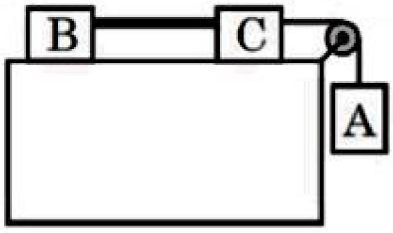
B) the reverse bias resistance of diode is very high in the beginning upto breakdown voltage is not achieved

C) both forward and reverse bias resistances are same for all voltages

D) both (A) and (B) are correct

For More Join: @IITJEE_Advanc

12. A block string system is shown in the figure. Mass of block A, B and C is $5\,kg$, $2kg$ and $2\,kg$ respectively. Mass of rod is given as $1\,kg$ and its length is $1m$. A transverse wave is transmitted in the rod in between the block B and C once in forward and once backward direction. Choose the **CORRECT** statements.



1) Time for string wave to reach one end to other end in forward direction is $\frac{2}{\sqrt{5}}\left(\sqrt{3}-\sqrt{2}\right)\sec$

2) Time for waves to reach one end to other end is same for both in forward and backward direction

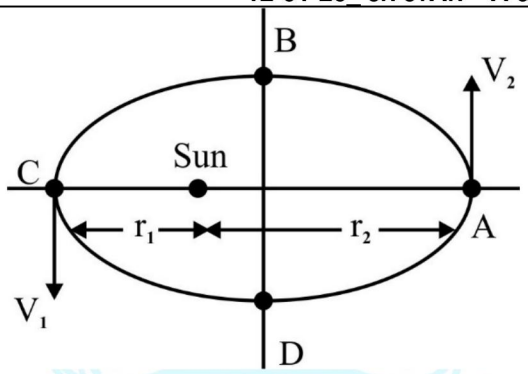
3) Time for wave to reach one end to other end is different for forward and backward direction.

4) Time for string wave to reach one to other end in backward direction si $\frac{1}{10}(15)^{1/2}\sec$

A) (1) and (2) both correct
B) Only (1) correct
C) (1) and (3) both correct
D) (2) and (4) both correct

13. A planet moves around the sun in an elliptical orbit as shown. Eccentricity of ellipse is $\frac{1}{2}$. Time taken by planet to move from D to B $\left(D\to A\to B\right)$ and B to $D\left(B\to C\to D\right)$ respectively are T_{DAB} and T_{BCD}.

For More Join: @IITJEE_Advanc

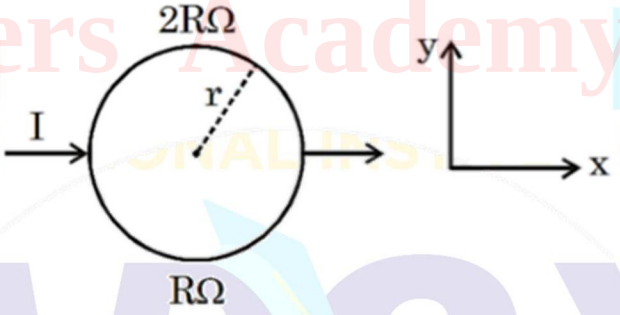


A) $\frac{T_{DAB}}{T_{BCD}}=\frac{\pi+2}{\pi-2}$ B) $\frac{T_{DAB}}{T_{BCD}}=\frac{2\pi+1}{2\pi-1}$

C) velocity of planet at point B is $\sqrt{3}v_2$

D) velocity of planet at point B is $\sqrt{2}v_2$

14. Two wires are bent (shown as joint-circle) with radius r (in xy plane). The upper half has resistance of $2R\ \Omega$ and the lower half of $R\ \Omega$. A current I is passed into circle as shown. The magnetic field at centre is



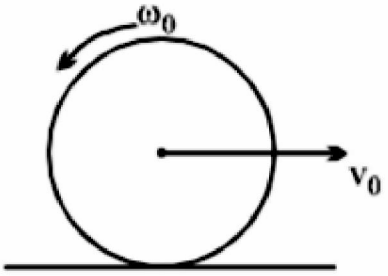
A) $\frac{\mu_0 I}{r}\hat{z}$ B) $\frac{\mu_0 I}{2r}\hat{z}$ C) zero D) $\frac{\mu_0 I}{12r}\hat{z}$

15. In a book, the answer for a particular question is expressed as $b=\frac{ma}{k}\left[\sqrt{1+\frac{2kl}{ma}}\right]$ here m represents mass, a represents acceleration, l represents length. The unit of b should be

A) m/s B) m/s^2 C) meter D) / sec

For More Join: @IITJEE_Advanc

16. A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is



A) $\frac{1}{2}$ B) 1 C) $\frac{3}{2}$ D) 2

17. An atom consists of three energy levels given by ground state with energy $E_0 = 0$, the first excited state with energy $E_1 = K$ and the second excited state with energy $E_2 = 2K$ where $K > 0$. The atom is initially in the ground state. Light from a laser which emits photons with energy $1.5K$ is shined on the atom. Which of the following is correct?

A) The photons are absorbed, putting one atom in a state E_1 and one atom in a state E_2.

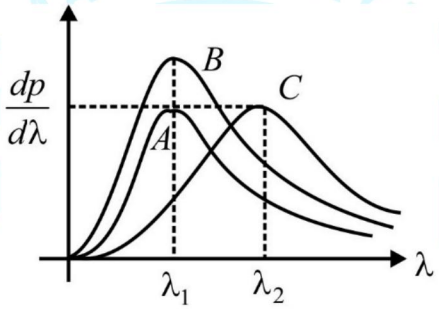
B) A photon will always be absorbed, but half the time the atom will go into the state with energy K and the other half into the state with energy $2K$. In this way, energy will be conserved on the average.

C) The atom absorbs a photon, goes into the first excited state with energy K and emits a photon with energy $0.5K$ to conserve energy.

D) The atom does not absorb any photon and stays in the ground state.

For More Join: @IITJEE_Advanc

18. Three stars radiate energy such that their radiated power depends on wavelength of radiation emitted as shown below. They can be considered as black bodies. The area under graph for A & C are equal. Which of the following statement(s) are CORRECT? dP is power emitted by star from wavelength λ to $\lambda + d\lambda$. $\frac{dP}{d\lambda}$ is plotted against λ



A) Emissive power of $A =$ Emissive power of $B <$ Emissive power of C

B) Temperature of $A <$ Temperature of $B =$ Temperature of C

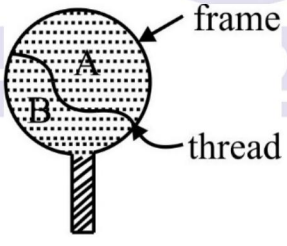
C) Temperature of $A =$ Temperature of $C <$ Temperature of B

D) Radius of $A <$ Radius of B; Radius of $C >$ Radius of A

19. An ideal diatomic gas goes through a cycle consisting of two isochoric and two isobaric lines. The absolute temperature of the gas rises 5 times both in the isochoric heating and in the isobaric expansion. Efficiency of such a cycle is:

A) 0.1 B) 0.2 C) 0.3 D) 0.4

20. A thread is tied slightly loose to a wire frame as in figure and the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A is heated, the thread



A) Becomes concave towards A B) Becomes convex towards A

C) Remains in the initial position

D) Either (1) or (2) depending on the size of A w.r.t B

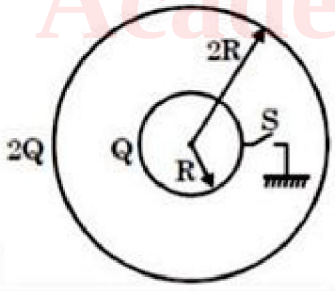
For More Join: @IITJEE_Advanc

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A particle of mass $5\times10^{-5}kg$ is placed at the lowest point of a smooth parabola having the equation $20x^2 = y\left(x, y \text{ in } m\right)$. Here y is the vertical height. If the particle is displaced slightly and is constrained to move along the parabola, then find the angular frequency (in rad/s) of small oscillations.

22. Two conducting spherical shell of radii R & $2R$ are given charges Q and $2Q$ respectively. Inner shell is provided with a switch which can ground the inner shell, as shown. Switch is initially open and energy stored in the system is U_1. After the switch is closed, energy stored in the system is found to be U_2. Find $\dfrac{U_1}{U_2}$



23. A locomotive of mass m starts moving so that its velocity varies according to the law $v = k\sqrt{s}$ where k is a constant and s is the distance covered. The total work performed by all the forces, which are acting on the locomotive during the first t seconds after the beginning of motion is found to be $\dfrac{mk^4t^2}{\beta}$. Find the value of β.

For More Join: @IITJEE_Advanc

24. If a galvanometer is converted into voltmeter with help of 32Ω resistance then voltage range becomes n times of initial. If same galvanometer is converted into ammeter with help of 2Ω resistance (as a shunt) then current range becomes n times of initial. Find resistance (in Ω) of original galvanometer.

25. A nucleus with atomic number, $Z = 72$ emits the following particles in a sequence: $\alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$. The Z of the resulting nucleus is $10N$. Then find the value of N

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. For the process: $He_{(g)} + Ne_{(g)} \rightarrow$ Gas mixture, at $T = 298K$ and 1 atmosphere pressure, the correct choice by assuming ideal behaviour is

A) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} = 0$ B) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} < 0$

C) $\Delta S_{system} = 0$ and $\Delta S_{surroundings} > 0$ D) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} = 0$

27. Which of the following pairs have same hybridization about the central atom?

A) SF_4 ; SiF_4 B) NO_3^- ; NF_3

C) SO_3 and SO_2 D) BrF_5 ; IF_7

28. Consider the following four aqueous solutions.

Solution I: 6% (w/w) urea solution.

Solution II: 0.585% (w/w) NaCl solution.

Solution III: 18% (w/w) glucose solution.

Solution IV: 34.2% (w/w) sucrose solution.

Which of the following statements is incorrect?

A) Boiling point of solution IV is greatest among all given solutions

B) Freezing point of solution II is greatest among all given solutions

C) Freezing point of solution I is more than solution IV

D) Boiling point of solution I is more than solution IV

For More Join: @IITJEE_Advanc

29. When 0.1 mol of $CH_3NH_2\left(K_b = 5\times10^{-4}\right)$ is mixed with 0.08mol of HCl and diluted to 1.0L, what will be the $\left[H^+\right]$ concentration in the solution

A) 1.25×10^{-4} B) $8\times10^{-11}M$ C) 1.6×10^{-11} D) $8\times10^{-2}M$

30. **Assertion:** Pentan-1-ol reacts with HBr to give a mixture of 1- and 2-bromopentanes in comparable amounts.

Reason: Simple 1^0 alcohols undergo substitution reaction with HBr by S_N2 mechanism.

A) Assertion is true, reason is true and is a correct explanation

B) Assertion is true, reason is true and is not a correct explanation

C) Assertion is true, reason is false

D) Assertion is false, reason is true

31. Which of the following statements about lattice enthalpy is/are correct?

(i) Lattice enthalpy is the energy required to completely separate one mole of an ionic solid into gaseous ions.

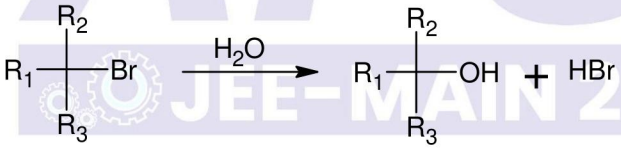
(ii) The crystal geometry of ionic compounds does not influence their lattice enthalpy.

(iii) It is not possible to calculate lattice enthalpy directly from the interaction of forces of attraction and repulsion only.

(iv) The stability of an ionic compound depends on its enthalpy of lattice formation.

A) i, ii, iv only B) i, ii, iii only C) ii, iii, iv only D) i, ii, iii, iv

32. The hydrolysis of the following alkyl halide is fastest when



A) All R_1, R_2, R_3 are hydrogens

B) R_2 is methyl and R_1,R_3 are phenyl groups

C) Only R1 is phenyl and other are hydrogens

D) All R_1, R_2, R_3 are methyl groups

For More Join: @IITJEE_Advanc

33. **STATEMENT – I:** Gibb's energy helps in predicting the best temperature and pressure conditions for industrial chemical processes to maximize the rate and efficiency.

STATEMENT – II: Gibbs energy changes in biochemical reactions are crucial for understanding cellular metabolism and energy production in living organisms.

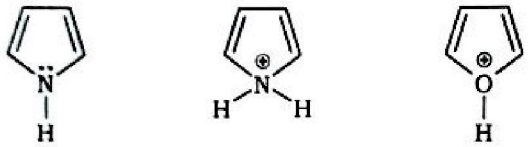
A) Both STATEMENT – I and STATEMENT – II are true

B) Both STATEMENT – I and STATEMENT – II are false

C) STATEMENT – I is true and STATEMENT – II is false

D) STATEMENT – I is false and STATEMENT – II is true

34.



(i) (ii) (iii)

The correct order of decreasing acidity is

A) ii > i > iii B) i > iii > ii C) iii > ii > i D) i > ii > iii

35. Three separate samples of a solution of a single salt is subjected for different tests and following observations are made.

Sample 1: A white precipitate is formed with excess of ammonia solution.

Sample 2: A white precipitate is observed with dil. NaCl solution.

Sample 3: A black precipitate is observed on passing H_2S.

The salt could be____

A) $AgNO_3$ B) $Pb(NO_3)_2$ C) $Hg(NO_3)_2$ D) $MnSO_4$

36. **Statement-1**: Metal carbonyls such as tetracarbonylnickel(0) and hexacarbonylchromium(0) are neutral complexes.

Statement-2: In metal carbonyls CO acts as σ donor and π acceptor ligand.

A) Both Statement 1 and Statement 2 are correct

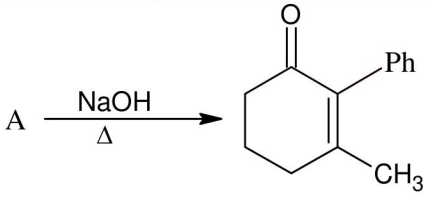
B) Both Statement 1 and Statement 2 are incorrect

C) Statement 1 is incorrect, but Statement 2 is correct.

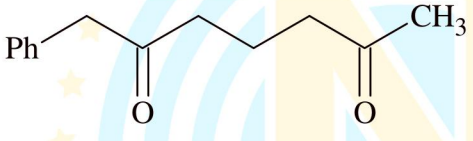
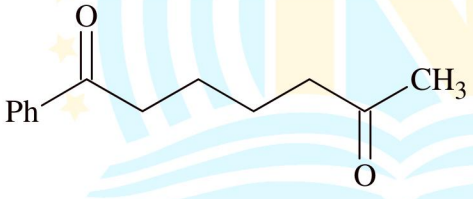
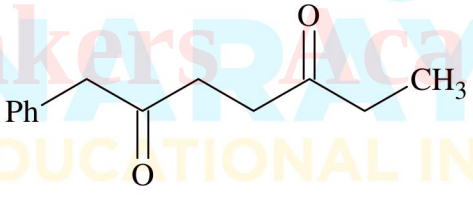
D) Statement 1 is correct, but Statement 2 is incorrect.

For More Join: @IITJEE_Advanc

37.



The reactant 'A' in the above reaction is?

A)	
B)	
C)	
D)	

38. Find the standard cell potential involving the cell reaction: $In^{2+} + Cu^{2+} \longrightarrow In^{3+} + Cu$, at 298 K

Given: $E^0_{Cu^{2+}/Cu} = x_1$ V; $E^o_{In^{3+}/In^+} = x_2$ V; $E^0_{In^{2+}/In^+} = x_3$V

A) $x_1 + x_3 - x_2$

B) $(x_1 + x_3 - 2x_2) / 3$

C) $x_1 + x_3 - 2x_2$

D) $x_1 + x_3 + 2x_2$

For More Join: @IITJEE_Advanc

39. For a gaseous reaction, $2A+B_2 \longrightarrow 2AB$, the following rate data was obtained at 300K.

Rate of disappearance of B_2. (mole litre^{-1}min^{-1})	[A] (mole/litre)	[B_2] (mole/litre)
1.8×10^{-3}	0.015	0.15
1.08×10^{-2}	0.09	0.15
5.4×10^{-3}	0.015	0.45

Calculate the rate of formation of AB when A is [0.02] and [B_2] is [0.04] mole litre^{-1} at 300K?

A) 1.28×10^{-3} M min^{-1} B) 2.56×10^{-3} M min^{-1}

C) 6.4×10^{-4} M min^{-1} D) 1.28×10^{-2} M min^{-1}

40. Find the **incorrect** statement about stereoisomerism?

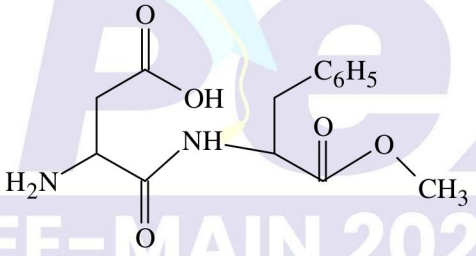
A) All chiral centers are stereogenic centers

B) All stereogenic centers are not chiral centerers

C) A compound may be chiral without chiral center

D) A compound can be chiral only if it has atleast one chiral center

41. Aspartame, an artificial sweetener is a peptide and has the following structure. Which of the following is correct about the molecule?



A) It has four functional groups and on hydrolysis produces only one amino acid

B) It has three functional groups and on hydrolysis produces only one amino acid

C) It has four functional group and on hydrolysis it produces a mixture of amino acids

For More Join: @IITJEE_Advanc

D) It has five functional groups and on hydrolysis it produces a mixture of amino acids.

42. Given below are two statements:

Statement- I: Oxygen being the first member of group 16 exhibits only -2 oxidation state.

Statement- II: Down the group 16 stability of +4 oxidation state decreases and +6 oxidation state increases.

In the light of the above statements, choose the most appropriate answer from the options given below.

A) Statement I is correct but Statement II is incorrect

B) Both the Statements I and II are correct

C) Both the Statements I and II are incorrect

D) Statement I is incorrect but Statement II is correct

43. An amine C_3H_9N reacts with Hinsberg's reagent to give a product which is insoluble in aqueous KOH. The amine is

A) $C_2H_5 — CH_2 — NH_2$

B) $CH_3 — CH(CH_3) — CH_2 — NH_2$

C) $C_2H_5 — NH — CH_3$ D) $(CH_3)_3 N$.

44 An alkene on reductive ozonolysis produces glyoxal (CHO – CHO) and methanal. The probable structure of alkene is

A) $CH_3 – CH = CH – CH = CH_2$ B) $CH_2 = CH – CH = CH_2$

C) $CH_3 – C \equiv C – CH_3$ D) $CH_2 = CH – C \equiv CH$

For More Join: @IITJEE_Advanc

45. Match the following column – I with column - II

Column – I	Column – II
P) $\left[Pt(gly)_2\right]$	1) Geometrical isomerism
Q) $\left[Co(en)_2Cl_2\right]Cl$	2) Optical isomerism
R) $\left[Cr(NH_3)_3(Py)_2(Br)\right]SO_4$	3) Linkage isomerism
S) $\left[Pt(NH_3)(PH_3)Cl(SCN)\right]$	4) Ionisation isomerism

A) P – 1 ; Q – 1, 2 ; R – 2, 4 ; S – 2,3

B) P – 1 ; Q – 1, 2 ; R – 1, 4 ; S – 1,3

C) P – 1, 2 ; Q – 1, 3 ; R – 1, 4 ; S – 1,3

D) P – 2 ; Q – 2, 4 ; R – 1, 4 ; S – 2,3

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. $POCl_3$ hydrolyses in water to give H_3PO_4 and HCl only. What volume in ml of 0.2M $Ca(OH)_2$ is needed to completely neutralize 50 ml 0.1 M $POCl_3$ solution.

47. A light of wavelength λ strikes a metal surface and ejects photoelectrons with $KE = 1eV$. Another light of wavelength $\frac{\lambda}{3}$ strikes the same metal surface and ejects photoelectrons with $KE = 4eV$. If the work function in eV is 'x', find the value of '10x'?

For More Join: @IITJEE_Advanc

48. How many of given statements are true for lanthanides (Ce to Lu)?

i) The common oxidation state of all lanthanides is +3.

ii) All lantanides belongs to 3rd group of modern periodic table.

iii) All lanthanides belong to seventh period of periodic table.

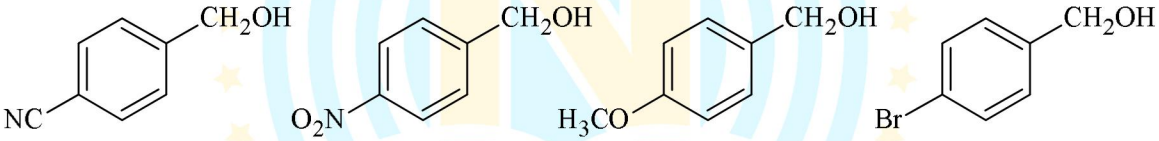
iv) The radius for all lanthanides cations(M^{+3}) decrease from Ce^{+3} to Lu^{+3}.

v) The basic character of hydroxide increases on moving from $Ce(OH)_3$ to $Lu(OH)_3$.

vi) All lanthanides are metals.

vii) Lanthanides react with water to liberate H_2.

49. How many of the following alcohols are less reactive than benzyl alcohol towards substitution with HBr?



I II III IV

50. Mole fraction of solute (AB) in an aqueous solution is 1/n. If 50% of AB dissociates into A and B and remaining 50% gets dimerised to $(AB)_2$, the new mole fraction of solvent becomes 4/5. The value of n is_____

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let f be a differentiable function satisfying $\int_0^x (x-p+1)f(p)\,dp = x^4 + x^2 \ \forall\ x \in [-2,\infty)$, then value of $\int_{-2}^{2} (f'(x)+f(x))\,dx$ is

A) 0 B) 36 C) 72 D) 18

For More Join: @IITJEE_Advanc

52. If $A=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\sin\left(\frac{r}{n^2}\right)$ and $B=\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{n}\sin\left(\frac{\pi r}{n}\right)$, then the value of $(2A+B)$ is

A) $1+\frac{2}{\pi}$ B) $\frac{2}{\pi}$ C) $\frac{\pi-1}{2}$ D) $\frac{\pi}{2}-1$

53. Let $f(x)$ be an even periodic function with period 4 defined partially as $f(x)=\begin{cases}e^x & ;\ x\in[0,1)\\ e(2-x) & ;\ x\in[1,2]\end{cases}$. In the interval $x\in[-10,10]$, the correct statement is

A) $f(x)$ is non-derivable at exactly 14 points

B) $f(x)$ is non-derivable at exactly 19 points

C) Range of $f(x)$ is $[0,1]$

D) $f(x)$ is discontinuous at exactly 8 points

54. If the function $f:R\to R$ is defined by $f(x)=|x|(x-\sin x)$, then which of the following statement is TRUE?

A) f is one-one but not onto B) f is onto but not one-one

C) f is both one-one and onto D) f is neither one-one nor onto

55. Water is being filled at the rate of 100 litre/min in a conical vessel (vertex downwards) with axis vertical. Its semi-vertical angle is $\tan^{-1}\left(\frac{5}{12}\right)$. The rate (in square meter per minute) at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 3 meters, is

A) $\frac{13}{75}$ B) $\frac{13}{25}$ C) $\frac{1}{5}$ D) $\frac{3}{11}$

56. The area of the region bounded by $A=\{(x,y):x^2-4\le y\le\min\{2x+11,9,12-6x\}$ is

For More Join: @IITJEE_Advanc

A) $\frac{511}{12}$ B) $\frac{947}{12}$ C) $\frac{437}{8}$ D) $\frac{240}{7}$

57. The true set of values of 'p' for which abscissa of all the points of extremum of the function $f(x)=x^3-3px^2+3(p^2-1)x+1$ lie in the interval $(-4,2)$, is

A) $(-3,5)$ B) $(-3,3)$ C) $(-1,3)$ D) $(-3,1)$

58. If the solution curve of the differential equation $\frac{dy}{dx}=\frac{x+y-4}{x-y}$ passes through the point $(3,2)$ and $(p+2,3)$; $p>0$, then

A) $2\tan^{-1}\left(\frac{1}{p}\right)=\log_e(p^2+1)$ B) $\tan^{-1}\left(\frac{1}{p}\right)=\log_e(p^2+1)$

C) $2\tan^{-1}\left(\frac{1}{p+1}\right)=\log_e(p^2+2p+2)$

D) $2\tan^{-1}\left(\frac{1}{p}\right)=\log_e\left(\frac{p^2+1}{p}\right)$

59. $\sum_{r=0}^{\infty}\tan^{-1}\left(\frac{r((r+1)!)}{(r+1)+((r+1)!)^2}\right)$ is equal to

A) $\frac{\pi}{2}$ B) $\frac{\pi}{4}$ C) $\cot^{-1}3$ D) $\tan^{-1}2$

60. If $f(t)=\int_1^t\frac{\log_{10}x}{1+x}dx$; $t>0$, then $f(e^4)+f(e^{-4})$ is equal to

A) 8 B) $\frac{16}{\log_e 10}$ C) $\frac{8}{\log_e 10}$ D) $\frac{e^4-1}{2\log_e 10}$

61. If a triangle is formed by the lines whose combined equation is given by $(x+y-4)(xy-2x-y+2)=0$, then the equation of its circumcircle is

For More Join: @IITJEE_Advanc

A) $x^2+y^2-5x-3y+8=0$ B) $x^2+y^2-3x-5y+8=0$

C) $x^2+y^2-3x-5y-8=0$ D) $x^2+y^2-5x-3y-8=0$

62. Let E_1 and E_2 be two ellipse whose centres are at origin. The major axes of E_1 and E_2 lie along x-axis and y-axis respectively. Let S be the circle $x^2+(y-1)^2=2$ such that the straight line $x+y=3$ touches the curves S, E_1 and E_2 at P,Q and R respectively, where $PQ=PR=\frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2 respectively, then

A) $e_1^2+e_2^2=\frac{43}{40}$ B) $e_1e_2=\frac{\sqrt{7}}{2\sqrt{5}}$

C) $\left|e_1^2-e_2^2\right|=\frac{5}{8}$ D) $e_1e_2=\frac{\sqrt{3}}{4}$

63. Let p and q be the longest and the shortest distance respectively of the point $(-7,2)$ from any point (α,β) on the curve whose equation is $x^2+y^2-10x-14y-51=0$, then geometric mean of p and q is equal to

A) 13 B) $5\sqrt{5}$ C) $2\sqrt{11}$ D) $3\sqrt{7}$

64. Through the point $(13,31)$ a straight line is drawn to meet the axes of X and Y at Q and S respectively. If the rectangle $OQRS$ is completed (where O is origin), then the co-ordinates of R satisfy the equation

A) $\frac{13}{x}+\frac{31}{y}=1$ B) $\frac{31}{x}+\frac{13}{y}=1$ C) $\frac{13}{x}-\frac{31}{y}=1$ D) $\frac{31}{x}-\frac{13}{y}=1$

65. A bag contains a large number of white and black marbles in equal proportions. Two samples of 5 marbles are selected (with replacement) at random. The probability that the first sample contains exactly 1 black marble, and the second sample contains exactly 3 black marbles, is

A) $\frac{25}{512}$ B) $\frac{15}{32}$ C) $\frac{15}{1024}$ D) $\frac{35}{256}$

For More Join: @IITJEE_Advanc

66. If z is a point on the Argand plane such that $|z-1|=1$, then $\frac{z-2}{z}$ is equal to

A) $\tan(\arg z)$ B) $\cot(\arg z)$ C) $i\tan(\arg z)$ D) $i\cot(\arg z)$

67. Let p,q and r are non-zero real numbers such that the system of equations

$(p+a)x+by+cz=0$

$ax+(q+b)y+cz=0$

$ax+by+(r+c)z=0$

has non-trivial solution, then the value of $\left(\frac{a}{p}+\frac{b}{q}+\frac{c}{r}\right)$ is

A) -1 B) 0 C) 1 D) 2

68. If $0 \le \alpha < \beta < \gamma \le 2\pi$ and if $f(x)=\cos(x+\alpha)+\cos(x+\beta)+\cos(x+\gamma)=0$, $\forall x \in R$, then value of $\gamma-\alpha$ is equal to

A) $\frac{2\pi}{3}$ B) $\frac{4\pi}{3}$ C) $\frac{\pi}{2}$ D) $\frac{\pi}{3}$

69. If $\vec{d}=\lambda(\vec{a}\times\vec{b})+\mu(\vec{b}\times\vec{c})+\gamma(\vec{c}\times\vec{a})$ and $[a\ b\ c]=1$, then $(\lambda+\mu+\gamma)$ is

A) $\vec{d}.(\vec{a}+\vec{b}+\vec{c})$ B) $\vec{a}+\vec{b}+\vec{c}$

C) 1 D) $2(\vec{a}+\vec{b}+\vec{c})$

70. A contest consists of predicting the results (win, draw or defeat) of 10 football matches. The probability that one entry contains at least 5 correct answers is

A) $\frac{9585}{3^{10}}$ B) $\frac{12385}{3^{10}}$ C) $\frac{9385}{3^{10}}$ D) $\frac{12585}{3^{10}}$

For More Join: @IITJEE_Advanc

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. If a is real constant and A,B,C are variable angles and $\left(\sqrt{a^2-4}\right)TanA+aTanB+\left(\sqrt{a^2+4}\right)TanC=6a$, then the least value of $Tan^2A+Tan^2B+Tan^2C$ is____

72. 5 boys $\left(B_1,B_2,B_3,B_4 \text{ and } B_5\right)$ and 5 girls $\left(G_1,G_2,G_3,G_4 \text{ and } G_5\right)$ are to be seated on a round table such that boys and girls sit alternately and B_i does not sit beside $G_i\ \forall i\in\{1,2,3,4,5\}$. If the number of such arrangements is N, then the sum of digits of N is equal to

73. A sample of 20 observations has mean of 50 and variance of 1, while another sample of 40 observations has mean of 50 and standard deviation 2. The two samples are combined to give complete set of 60 observations with variance σ^2, then $3\sigma^2$ is equal to

74. Let $a=\left(4^{1/401}-1\right)$ and $b_n={}^nC_1+{}^nC_2.a+{}^nC_3.a^2+.....+{}^nC_n.a^{n-1}$, such that $b_{2006}-b_{2005}=x^y$; where x and y are co-prime naturals, then xy is equal to

75. Let S_n denotes the sum of infinite terms of a GP whose first term is $\left(n^2-1\right)$ and common ratio is $\frac{1}{n}\ \forall n\in\{1,2,3,.....\}$, then the value of $\sum_{n=1}^{\infty}\frac{S_n}{2^{n-1}}$ is

For More Join: @IITJEE_Advanc